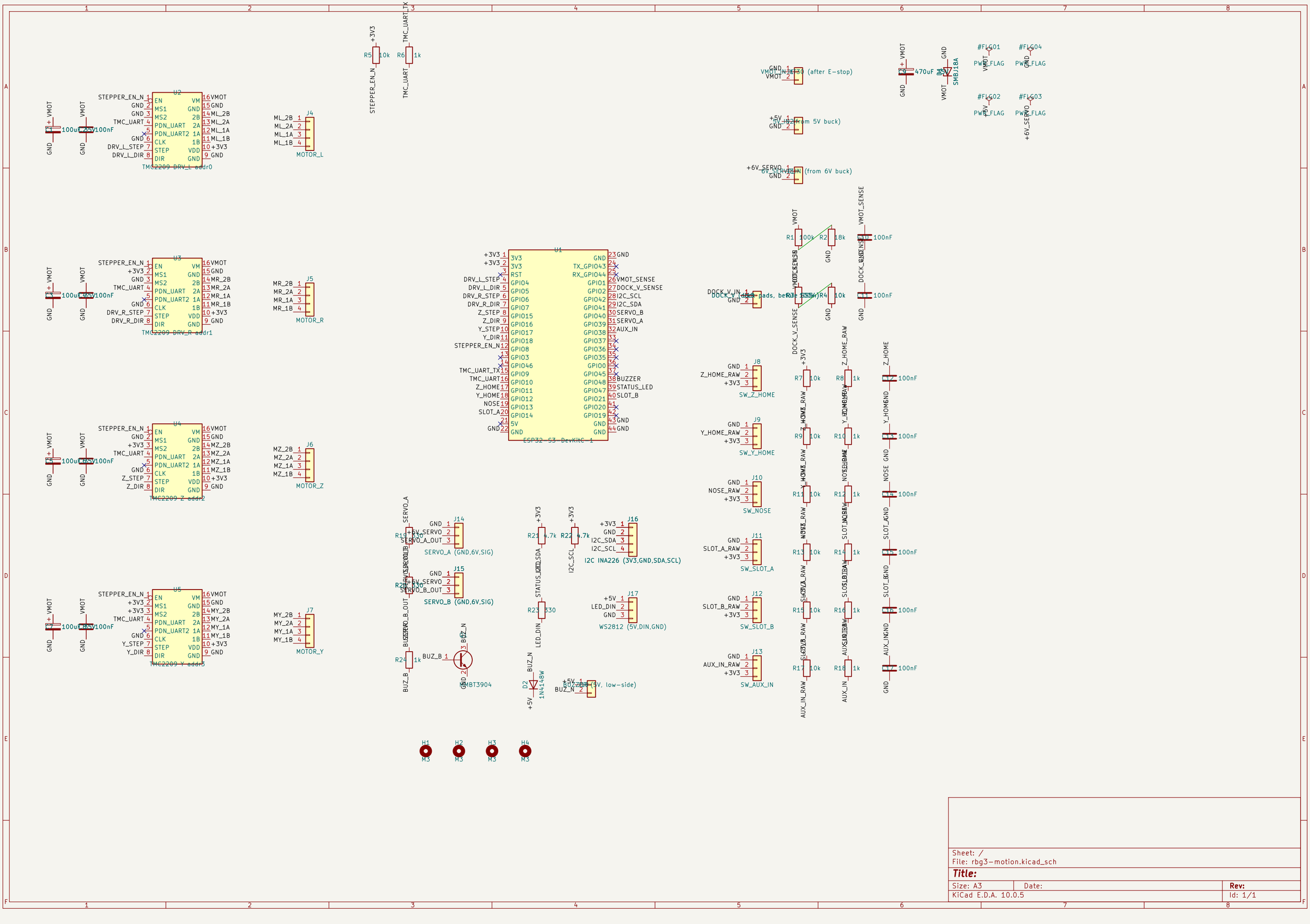

U2
TMC2209_DRV_L_addr0
U3
TMC2209_DRV_R_addr1
U4
TMC2209_Z_addr2
U5
TMC2209_Y_addr3
U1
ESP32-S3-DevKitC-1
J4
MOTOR_L
J5
MOTOR_R
J6
MOTOR_Z
J7
MOTOR_Y
J8
SW_Z_HOME
J9
SW_Y_HOME
J10
SW_NOSE
J11
SW_SLOT_A
J12
SW_SLOT_B
J13
SW_AUX_IN
J14
SERVO_A (GND,6V,SIG)
J15
SERVO_B (GND,6V,SIG)
J16
I2C INA226 (3V3,GND,SDA,SCL)
J17
WS2812 (5V,DIN,GND)
H1
H2
H3
H4
M3
VMOT
GND
+3V3
+5V
+6V_SERVO
STEPPER_EN_N
TMC_UART
TMC_UART_TX
DRV_L_STEP
DRV_L_DIR
DRV_R_STEP
DRV_R_DIR
Z_STEP
Z_DIR
Y_STEP
Y_DIR
ML_2B
ML_2A
ML_1A
ML_1B
MR_2B
MR_2A
MR_1A
MR_1B
MZ_2B
MZ_2A
MZ_1A
MZ_1B
MY_2B
MY_2A
MY_1A
MY_1B
R5 10k
R6 1k
VMOT_IN (after E-stop)
+5V (from 5V buck)
+6V_SERVO (from 6V buck)
470uF 35V
SMBJ18A
#FLG01
#FLG02
#FLG03
#FLG04
PWR_FLAG
R1 100k
R2 18k
R3 10k
R4 10k
100nF
DOCK_V_SENSE
DOCK_V_IN (pads, before R1/R2)
VMOT_SENSE
R7 10k
R8 1k
Z_HOME_RAW
Y_HOME_RAW
NOSE_RAW
SLOT_A_RAW
SLOT_B_RAW
AUX_IN_RAW
Z_HOME
Y_HOME
NOSE
SLOT_A
SLOT_B
AUX_IN
R9 10k
R10 1k
R11 10k
R12 1k
R13 10k
R14 1k
R15 10k
R16 1k
R17 10k
R18 1k
R21 4.7k
R22 4.7k
I2C_SCL
I2C_SDA
STATUS_LED
LED_DIN
R23 330
R24 1k
BUZ_B
BUZ_N
Q2 2N3904
D2 1N4148W
BUZZER (5V, low-side)
SERVO_A
SERVO_B
SERVO_A_OUT
SERVO_B_OUT
R19 330
R20 330
100uF 35V
Sheet: /
File: rbg3-motion.kicad_sch
Title:
Size: A3
Date:
Rev:
KiCad E.D.A. 10.0.5
Id: 1/1